

Anil Seth



Safer Driving Using Autonomous Cars in India

Autonomous cars have evoked keen interest because of huge investments announced by the Obama Administration. Here are a few projects that could be undertaken by students and researchers in this area. These projects are very practical and would hone the skills of software developers.

You can imagine the pace at which self-driving or autonomous cars are progressing from the recent announcement of an investment of US\$ 4 billion by the Obama administration to help make them a reality. A guiding principle for the administration is that “it has an interest in cutting the death toll” (see the Slashdot news item <http://goo.gl/JL5oBb>). However, self-driven cars seem like an impossible challenge on Indian roads.

This is an area I would have loved to explore if I were a student today. A combination of Raspberry Pi and Arduino makes it possible to program and experiment with complex ideas and tasks even in a college environment.

Just try to imagine how the other vehicles on an Indian road would react if an autonomous car tried to parallel park, knowing that self-driving cars are modelled on polite drivers! So, you need projects which are specifically designed and feasible for Indian conditions.

Autonomous headlights

A solution to the excessive use of high beams in Delhi many years ago was to cover the upper half of the headlights with a black tape. The traffic police just could not get drivers to use low beams while in the city. However, if every headlight was autonomous, it would sense oncoming traffic and switch to low beam without any action needed from the driver.

It would definitely be a fun project to implement on toy cars, along with your friends from electronics and mechanical engineering. To know its utility, just ask your grandparents!

Speed breakers and pot holes

You may have experienced the sudden creation of a speed breaker even on a road with which you are very familiar. On unfamiliar roads, speed breakers are far too often a surprise. Even familiar small pot holes can surprise

you, and you may not be able to react fast enough. While the bump you may experience is *usually* not dangerous, it is uncomfortable.

So, another terrific project would be to identify speed breakers or hurdles on the road, especially in low light environments. The car should then over-ride the accelerator and autonomously slow down to a safe speed. You may even add an LED display at the back of the car to warn the vehicles behind you!

Where's the road?

It is embarrassing to admit but, a couple of times, I have tried to make space for an oncoming truck and slid off the narrow roads in Goa into a ditch! So, I would love for you to implement a project which detects the edge of the road, and steers the car safely between the fearsome truck and the ditch on the side.

This project can be useful in metros for tipsy drivers as well, and steer the car away from the side walk.

Stop signs and pedestrian crossings

A final project I would like you to consider is one where the autonomous car recognises a stop sign or a stop signal, and comes to a complete halt for at least one second. It should also slow down at a pedestrian crossing if there is no stop sign or signal there.

While this project may not be implemented in real life, it still could be a good learning tool for understanding vision.

Finally, software and technology has made our life much easier. It would be nice, now, to use it for making our life safer as well.  **END**

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